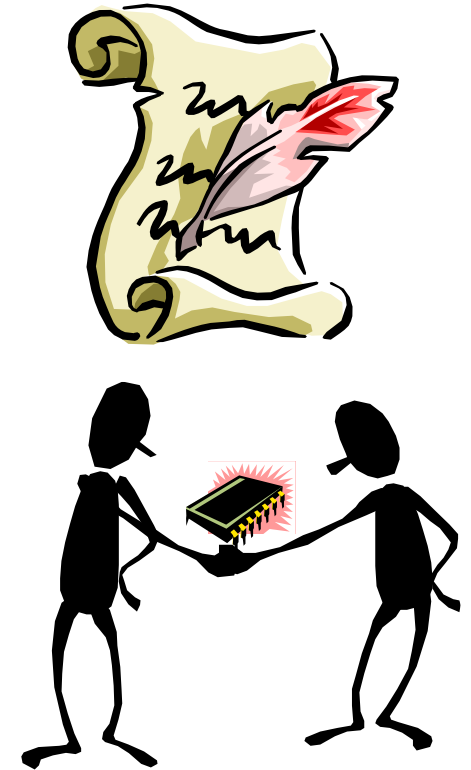


Applications

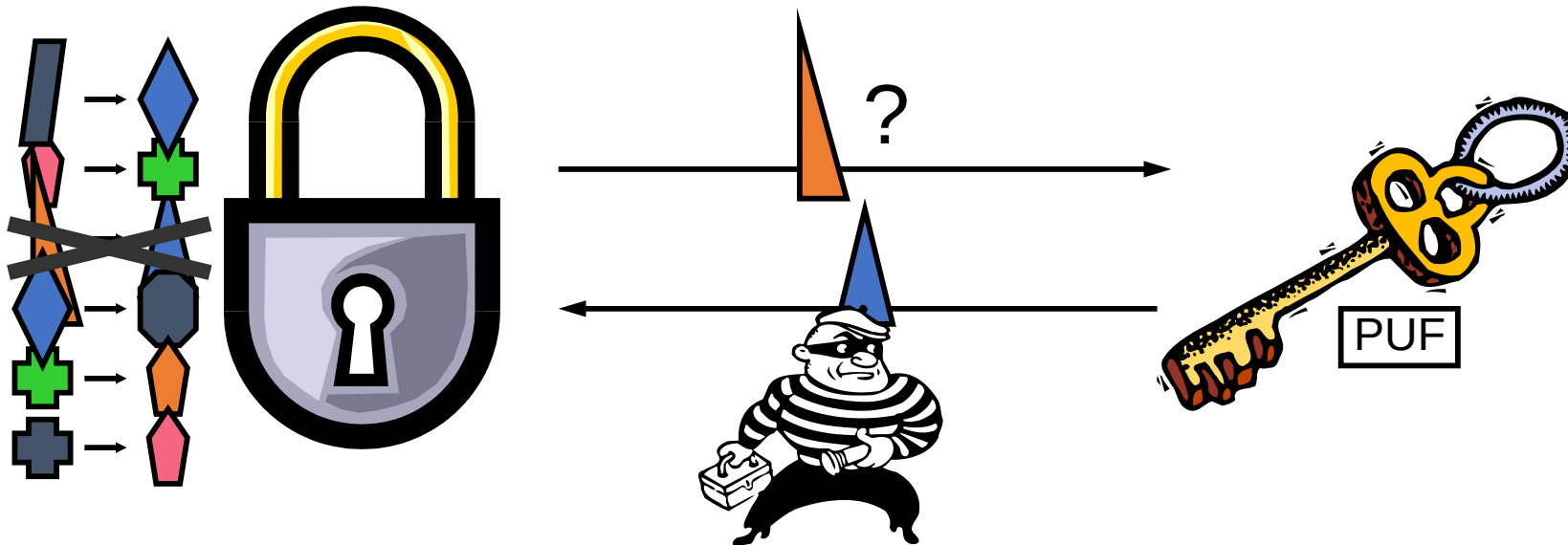
- Anonymous Computation
 - Alice wants to run computations on Bob's computer, and wants to make sure that she is getting correct results. A certificate is returned with her results to show that they were correctly executed.
- Software Licensing
 - Alice wants to sell Bob a program which will only run on Bob's chip (identified by a PUF). The program is copy-protected so it will not run on any other chip.



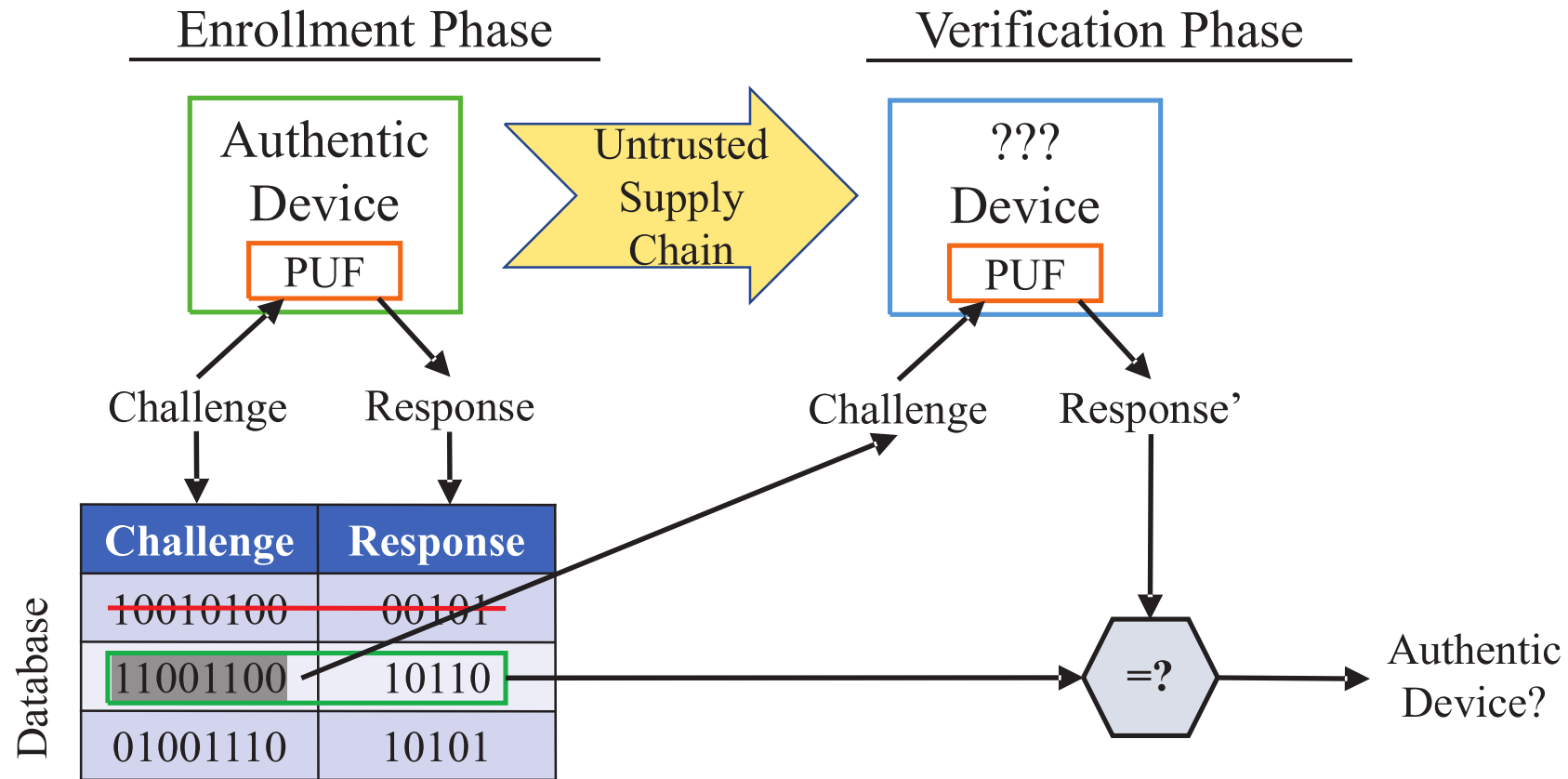
We can enable the above applications by trusting only a single-chip processor that contains a silicon PUF.

Using a PUF as an Unclonable Key

- A silicon PUF can be used as an unclonable key.
 - The lock has a database of challenge-response pairs.
 - To open the lock, the key has to show that it knows the response to one or more challenges.



Strong PUF-based Authentication



Key Generation with PUFs

